

Memory transfer and continual learning in predictive-coding associative networks

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Abstract

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Author summary

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1 Introduction

2 Model and analysis

2.1 A two-population predictive-coding memory

2.2 Student inference and learning descend a free energy

Proposition 1. [Student state relaxation is exact gradient descent on F_S ; the zero-diagonal weight update is projected gradient descent on F_S .]

2.3 The student steers the teacher up the student's surprise

Theorem 1. [Fast student state, frozen student weights. Isolating the student's contribution to the teacher's sleep dynamics — the push $u = \pi_{ST}\varepsilon_{TS}$ — the teacher performs exact gradient ascent on the student's settled variational free energy $F_S^{\text{eq}}(x_T) \equiv \min_{x_S} F_S(x_S, x_T)$: $\tau_T \dot{x}_T|_{\text{push}} = (|\pi_{ST}|/\pi_{TS}) \nabla_{x_T} F_S^{\text{eq}}(x_T)$ — at every point, with no approximation, hence locally steepest ascent of the student's surprise. Caveats: local ascent (not the global summit); the push rescales but does not seed an empty direction (noise seeds it).]

2.4 The landscape: the novelty operator

Lemma 1. [Fast-student steady state gives $x_S^* = (I - N_S)x_T$, hence $\varepsilon_{TS} = -N_S x_T$, with $N_S = \pi_S S_S (\pi_{TS} I + \pi_S S_S)^{-1}$; N_S is symmetric PSD and $N_S u = 0$ on learned directions.]

2.5 The surprise identity: the settled student's free energy is the novelty score

Corollary 1. [At the fast-student equilibrium, $F_S^{\text{eq}}(x_T) = \frac{\pi_{TS}}{2} x_T^\top N_S x_T$: the landscape Theorem 1 climbs is the novelty-weighted energy of the teacher's state — a quadratic that is flat on learned directions and curved on novel ones.]

Remark 1. [Axis by axis in the student's eigenframe, $\dot{c}_k = (|\pi_{ST}|/\tau_T) n_k c_k$: learned directions frozen, novel directions amplified in proportion to their novelty n_k ; the push on each direction extinguishes as the student learns it.]

2.6 Staying on the memory manifold

Lemma 2. *[Under the guard $|\pi_{ST}| < \pi_T \sigma_{\min}^2$, the reduced sleep drift’s quadratic form is strictly negative on the off-manifold subspace: $q^\top A q \leq -(\pi_T \sigma_{\min}^2 - |\pi_{ST}|) \|q\|^2 < 0$ for $0 \neq q \perp \ker M_T$ — on that subspace, damping beats drive. A quadratic-form bound, not manifold invariance or containment of a mixed state: containment in the full model (cross-block coupling, leash, noise) is verified numerically.]*

2.7 An exact free-energy saddle

Proposition 2. *[The physical functional $\Phi = F_T - F_S$ generates the sleep dynamics (on the teacher sphere, with the model’s mobilities and projected W_S ascent) iff $\pi_{ST} = -\pi_{TS}$ — by Theorem 1, exactly when the teacher’s ascent rate on the student’s surprise matches the precision π_{TS} that defines it. The teacher descends Φ ; the student, descending F_S (Proposition 1), ascends Φ : a minimax on one potential. Shape reminiscent of the active-inference agent–environment saddle, not identical.]*

2.8 Scope: subspace, not named episodes

Remark 2. *[Linearity $\Rightarrow$ transfer of a subspace / effective rank, not named patterns; episodic one-per-step replay needs an added nonlinearity.]*

3 Results

3.1 Prioritized transfer of unlearned directions

[figure placeholder]
Fig 1. Novelty-spectrum “staircase” and x_T -manifold alignment: the student acquires unlearned directions; “reliable completion across tested regimes” (not global-convergence). Steps are a timescale/plotting artifact.

3.2 The wake/sleep saddle picture

[figure placeholder]
Fig 2. Saddle cross-sections: a bowl along a predictable direction vs. a hill along a novel one; the exact $\Phi = F_T - F_S$ picture at $\pi_{ST} = -\pi_{TS}$.

3.3 Merging memories by interleaving

[figure placeholder]
Fig 3. Interleaved rehearsal merges two teachers: the crosstalk sawtooth decays to zero (union of memory subspaces), demonstrated in simulation.

3.4 Continual learning without catastrophic forgetting

[figure placeholder]
Fig 4. Continual learning: retention across a streamed sequence vs. a buffer-only control that catastrophically forgets.

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